

Limiting gizmos in a car

The U.S. Dept. of Transportation wants laws limiting the variety of gizmos that a manufacturer can install in a car

Tesla and Mercedes join hands

Tesla will provide the powertrain for an electric Mercedes set to be launched later this year

H.A.N.S. PROTECTING THE RACING DRIVER'S NECK



You thought speed kills. Well, actually, it's stopping instantly that's fatal. Safe to say, HANS has saved the necks of a lot of racing drivers over the last few years

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“**T**here are only three sports: bullfighting, motor racing and mountaineering; all the rest are merely games,” are the words of Ernest Hemmingway, American novelist and three-time winner of the Nobel Prize for Literature. The fact that he puts motorsports in the same category as bullfighting and mountaineering shows the very real danger the driver is in every time he straps himself with the five-point safety harness. And these dangers haven't gone away despite multiple safety measures.

After the fateful Imola Grand Prix weekend in 1994 that saw the demise of two drivers Ayrton Senna and Roland Ratzenberger, Formula 1 fans have been spared of any further fatal crashes. But there've been many dangerous ones over the years. Seven Indy Racing League and ChampCar drivers and four Nascar drivers (Adam Petty, Kenny Irwin, Tony Roper and Dale Earnhardt Sr.) have had fatal crashes.

What was common among all of them? They all had a fracture at the base of their skulls. The cracks happen when the head moves around freely upon impact, while the safety harnesses securely hold the rest of the body back. In simpler terms, it's essentially a whiplash.

What prompted HANS development?

While most people believe that the Head and Neck Safety (HANS) device came into being over the past 10 years, work on this began way back in 1981. Dr. Robert Hubbard was at the Uni-

versity of Michigan Highway Safety Research Institute, understanding the causes of head injuries during car accidents. What kickstarted this was the death of IMSA driver, Patrick Jacquemart during testing at the Ohio race track. Dr. Hubbard got in touch with his brother-in-law, Jim Downing, who was also a race-car driver and a close friend of Patrick. After understanding that Jacquemart had died because of what is now known as basilar skull fracture, the two teamed up and started working on a device that would keep the driver's neck secure at all times.

What HANS is meant to do

HANS is a U-shaped device that sits behind the neck, and has two shoulder supports on either side of the neck, extending all the way to the top of the chest. The critical bit of this device is the portion that starts at the base of the neck and extends half-way up the rear skull (an extender collar, if you will). When the racing driver wears his crash helmet, this portion is supposed to be outside the helmet, not inside. There are two tethers on either side of the HANS portion, which lock on to the helmet. In case of impact, these tethers are the most critical. By restricting sudden head movements, the energy (g-force, in other words) that would usually be in the neck and upper spine area is shifted to the front of the head, with very limited (padded or muffled, if you may) movements. With this in place, the head won't suddenly whip forwards and backwards in case of an impact. While the neck isn't held back stiffly, the idea is to keep the position relative to that of the body, preventing sudden and extreme